



UNIVERSITAS
GADJAH MADA

PHYLOGENETIC ANALYSIS OF THE BINTURONG, *Arctictis binturong* (Raffles, 1821), USING THE COI MARKER

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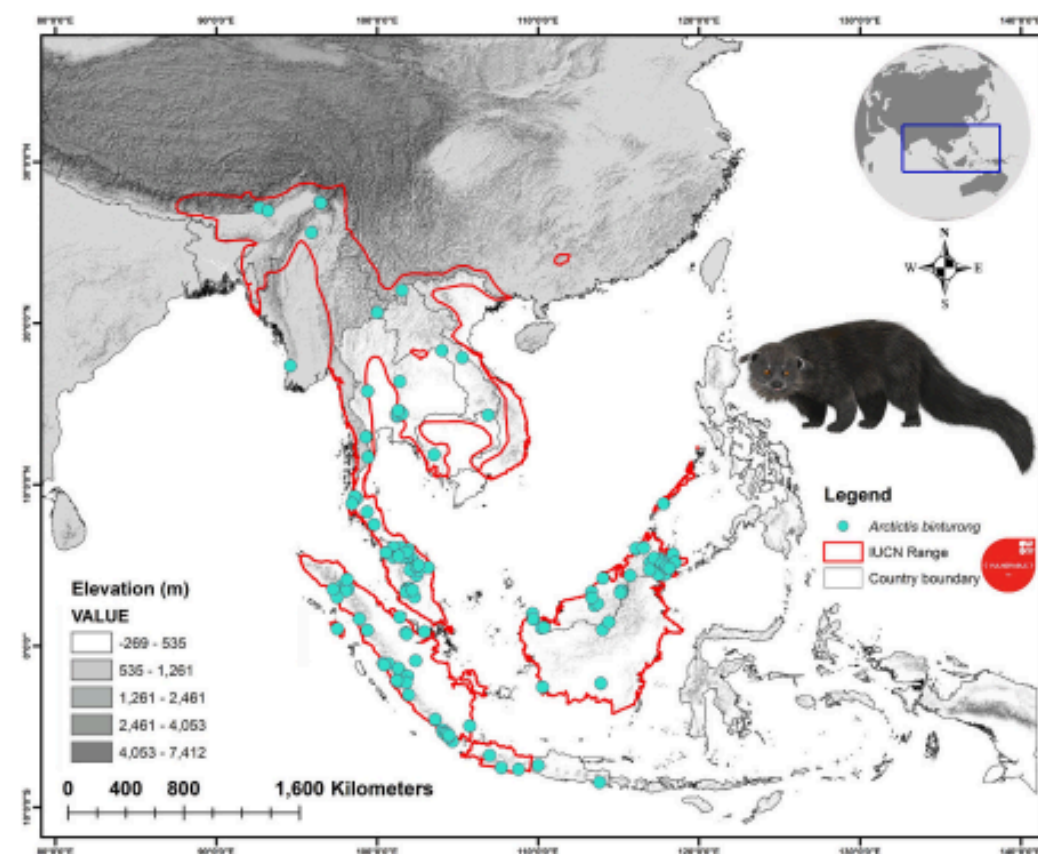


LOCALLY ROOTED, GLOBALLY RESPECTED



Introduction

- *Arctictis binturong* is the largest member of the family Viverridae and is widely distributed across South and Southeast Asia.
- Despite its broad distribution, binturong populations are threatened by habitat loss, forest fragmentation, hunting, and illegal wildlife trade.
- Binturong plays an important ecological role as an omnivorous frugivore and seed disperser in forest ecosystems.
- Due to ongoing population decline, the species is currently classified as **Vulnerable by the IUCN**.
- Molecular genetic information, particularly phylogenetic relationships, remains limited and requires further investigation.



the study area for the present study along with the IUCN extant of binturong. The figure also highlights the location poi

Source: IUCN Red List (2024)



Source : petto photo





Scientific Problem

Most existing studies on binturong focus on:

- ecology
- behavior
- ex-situ conservation

However, molecular genetic information remains limited

Why This Study?

Understanding genetic relationships is important for:

- ✓ species identification
- ✓ evolutionary relationship analysis
- ✓ conservation strategy development

Research Gap

Limited phylogenetic studies of *Arctictis binturong* using molecular markers, particularly COI.





Research Problems, Research Objectives, and Research Benefits

Research Problems

- How can COI marker be used for molecular identification of **Arctictis binturong**?
- How is the phylogenetic relationship among binturong samples based on COI sequences?

Research Benefits

- ✓ Analyze COI DNA sequences from binturong samples
- ✓ Reconstruct phylogenetic relationships among individual
- ✓ Evaluate the usefulness of COI marker for molecular identification

Research Objectives

- Provides molecular baseline data for binturong conservation
- Supports phylogenetic understanding of the species
- Contributes to future genetic conservation strategies



LITERATURE REVIEW

- Largest species within the family Viverridae (Kalita & Debroy, 2021)
- Classified as Vulnerable by the IUCN due to habitat loss and illegal wildlife trade (Anggraini et al., 2023)
- Molecular genetic information remains limited



Taxonomy

- Kingdom : Animalia
 - Filum : Chordata
 - Subfilum : Vertebrata
 - Classis : Mamalia
 - Ordo : Carnivora
 - Famili : Viverridae
 - Genus : *Arctictis* (Temminck, 1824)
 - Spesies : *Arctictis binturong* (Raffles, 1821)

- Body Coloration : Black to dark brown coarse fur, White-tipped ears and long whiskers, Provides camouflage in forest environments
- Morphological Adaptation: Prehensile tail for arboreal locomotion, Strong limbs and curved claws for climbing, Adapted to a nocturnal lifestyle



LITERATURE REVIEW

COI As a DNA Barcoding Marker

- mitochondrial DNA marker
- widely used in DNA barcoding
- suitable for species identification
- effective for phylogenetic reconstruction (Strüder-Kypke & Lynn, 2010).

Phylogenetics

Phylogenetics is the study of evolutionary relationships among organisms based on genetic similarity.

Phylogenetic trees are used to:

- determine genetic relationships
- estimate evolutionary divergence
- identify species relationships (Munjal *et al.*, 2019).

Mitochondrial DNA

Mitochondrial DNA is widely used in phylogenetic studies because:

- ✓ maternally inherited
- ✓ relatively high mutation rate
- ✓ suitable for species-level analysis (Patwardhan *et al.*, 2014)

Hypothesis

COI marker can be used effectively for molecular identification and phylogenetic analysis of *Arctictis binturong*.



RESEARCH METHODS

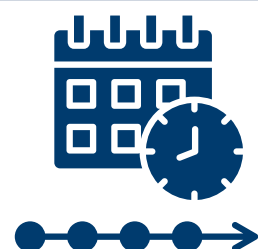
Study Design & Setting



STUDY DESIGN

Molecular
phylogenetic study

STUDY PERIOD



September 2026 –
Desember 2027

SAMPLE SOURCE



Balai Besar Konservasi
Sumber Daya Alam
(BKSDA), East Java

LABORATORY ANALYSIS

Laboratory of Sistematic
Animal of Faculty of Biology
UGM

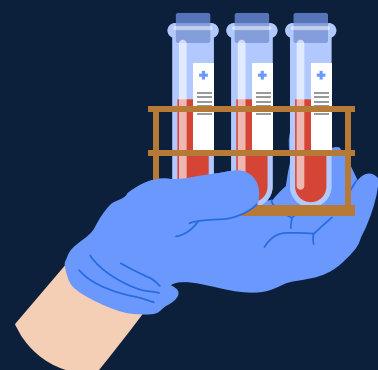
DNA SEQUENCING

Integrated Research and
Testing Laboratory (LPPT UGM)



RESEARCH METHODS

1



Sample Collection

Binturong
biological
samples:
8 hair samples
3 blood samples

2



DNA Extraction Using:

Qiagen DNeasy
Blood & Tissue Kit

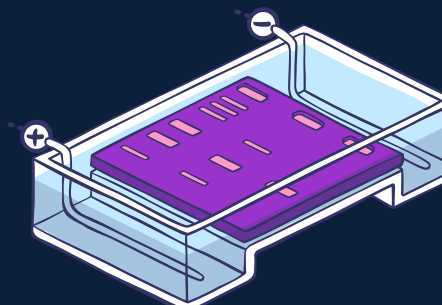
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PCR Amplification Target gene:

COI Marker

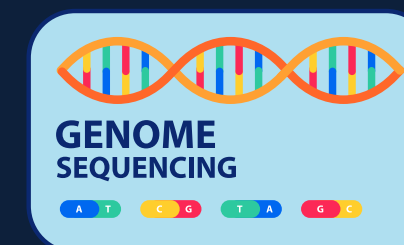
4



Electrophoresis

DNA fragment
visualization

5



DNA Sequencing

PCR products
sent for
sequencing

6



Bioinformatics Analysis

Sequence
identification and
phylogenetic
reconstruction



LABORATORY METHODS

DNA EXTRACTION

→ DNA extraction was performed using the Qiagen DNeasy Blood & Tissue Kit.

Reagents:

- ATL Buffer (180 μ L)
- Proteinase K (20 μ L)
- AL Buffer (200 μ L)
- Ethanol (200 μ L)
- AW1 & AW2 Buffer
- AE Buffer (70 μ L)

PCR AMPLIFICATION

→ PCR amplification of the COI gene was conducted using:

- KOD One PCR Master Mix (12.5 μ L)
- Forward primer (1 μ L)
- Reverse primer (1 μ L)
- ddH₂O (4.5 μ L)
- DNA template (6 μ L)

Total reaction volume:
25 μ L

ELECTROPHORESIS

→ Conditions:

- Agarose gel: 1.2%
- Voltage: 50 V
- Duration: 45 minutes

Successful amplification indicated by visible DNA bands.



BIOINFORMATICS ANALYSIS



1 Analyze COI DNA sequences from binturong samples

→ Compare COI sequences with GenBank database to confirm species identity.



2 Sequence Editing, GeneStudio Pro 2.2

→ Trim and edit DNA sequences to obtain high-quality consensus sequences.

3 Multiple Sequence Alignment MEGA 11 (ClustalW)

→ Align COI sequences to identify homologous nucleotide positions.



4 Phylogenetic Reconstruction IQ-TREE

→ Reconstruct phylogenetic relationships using Maximum Likelihood with 1000 bootstrap replicates.

5 Bayesian Analysis, BEAST 2.7.7

→ Estimate evolutionary divergence using Bayesian MCMC analysis.



6 Tree Visualization

→ Visualize and interpret phylogenetic tree topology and branch support values.



Research Timeline

No	Activity	Period
1	Initial research topic consultation and discussion with supervisor	January 2026
2	Research topic finalization and proposal framework development	March 2026
3	Proposal seminar writing and manuscript preparation	March- April 2026
4	Proposal submission and seminar presentation	June 2026
5	Laboratory work and field sample collection	September-November 2026
6	Bioinformatics analysis, data interpretation, and report writing	Desember- onwards



Key References

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Thank You

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